

CROSS-GAME NFTS/ INTERPLAY

Redefining Digital Ownership

Jessy Ssebuliba/Noah Tayebwa/Daniel Ntege

The Problem: You don't really own what you earn

- Today's gaming industry locks your items, skins, and rewards inside a single game's control
- Players lose their digital assets when they stop playing or when a game shuts down in which they invested both time and lots of money
- Game currencies and NFTs are isolated, non-interoperable, and centrally controlled.
- "Ownership" in gaming is currently an illusion — it's stored and controlled by a central governance.

CROSS-GAME NFTs ECOSYSTEM

The Blueprint for Cross-Game Digital Ownership. Further discussions to improve the concept are still needed

- **Cross-Game NFTs:** Items earned in one game should be recognized and usable in others — ownership lives in your wallet, not the publisher's servers.
- **Enhanced Metadata Standard:** Each NFT carries structured data (origin, category, compatibility, ecosystem) that allows any game to interpret it correctly for cross compatibility.
- **Decentralized Asset Storage:** All game assets and metadata are stored via IPFS, ensuring permanence, immutability, and maximum transparency.
- **Locking & Unlocking Mechanism:** NFTs can be temporarily locked when actively used (e.g., equipped in a tournament) to prevent transfers — a step toward ERC-6982-like standards on Hedera
- **Cross-Chain Vision:** Future iterations will allow NFTs to be bridged between blockchains — so a skin minted on Hedera could later appear in Ethereum, Polygon, or other networks.
- **Secure Transfers:** All transfers are on-chain and verifiable, eliminating central authority risk while maintaining interoperability across supported ecosystems.

Technology & How It Works

We created two games **Dino**, and **Pixel racer** to demo the idea. They're totally different games with different token ids on Hedera

- Built on Hedera Token Service (HTS) via a smart contract to support none EVM wallets like Metamask:
Used for minting, managing, and verifying NFTs with ultra-low fees, high throughput, and finality in seconds.
- Decentralized Storage via IPFS (PINATA):
Every NFT references immutable metadata and assets through IPFS CIDs, ensuring transparency and permanence.
- Interplay Metadata Schema:
The core of the idea at this stage are the few additions to the metadata that defines the nfts which are a sort of standard for both game.
origin_game (where the NFT was minted)
item_category (skin, theme, weapon, etc.)
compatibility (supported game IDs, versions, or platforms)
This enables universal interpretation across different games.
- Game Integration:
Dino mints new NFTs upon a new highscore
Pixel Racer reads the same NFTs from the player's wallet and applies them as car skins. Same the other way round
Both games verify ownership using the Hedera Mirror Node API.
- Locking & Unlocking Workflow
Future integration of an **NFT Lock Contract**, inspired by ERC-6982, will allow in-game "equipped" assets to be locked and prevent on-chain transfers until unequipped.
- Cross-Chain Bridge (Future Phase):
Planned extension enabling NFTs minted on Hedera to be **bridged to Ethereum or Polygon** using standardized metadata and bridging protocols.

Market & Opportunity

Redefining Africa's \$1.8 billion rapidly growing Industry

- The Web3 Gaming Gap:
Fewer than **1% of games** currently offer real asset ownership.
Most NFTs are **locked within ecosystems**, making them unusable elsewhere.
Players lose **lifetime value** when switching or abandoning games.
- Africa: The Emerging Powerhouse:
349 million players, majority on mobile — a massive, untapped market for interoperable gaming economies.
- Our Opportunity:
Create the first universal, game-agnostic NFT ownership layer on Hedera.
Enable studios to easily join an **interoperable ecosystem**, sharing player engagement and item value across titles
Position Interplay as the **standard bridge** between closed game economies and the new open digital world

Competitive Analysis

Overview of the underlying market state

Platform / Project	Focus Area	Limitations / Gaps	How Interplay Stands Apart
Axie Infinity	Play-to-Earn Game	Closed ecosystem, NFTs usable only within Axie	Interplay creates NFTs usable across games
Immutable X	Game Asset Marketplace	Strong for minting, weak for cross-game logic	Adds metadata standards + interoperability
Gala Games	Multi-game Publisher Ecosystem	Centralized; limited interoperability outside Gala	Decentralized, open, Hedera-based model
Traditional Studios (EA, Epic, etc.)	Closed asset systems	Assets owned and controlled by publishers	Players own verifiable, transferable assets

Business Model

Ownership creates value. How cross-game nfts generates value

- **NFT Utility & Marketplace Layer**
Cross-Game NFT Marketplace — Players can trade, sell, or rent assets recognized by multiple titles.
In-Game NFT Customization Fees — Skins and themes can be upgraded or fused, with micro-fees flowing back to the ecosystem.
Royalties — Built-in royalties on secondary market trades ensure continuous revenue for both developers and Interplay.

Team & Vision

- We bridge traditional game development and decentralized innovation — combining the polish of professional engineering with the spirit of open-source creativity.
- We envision a gaming world where players can earn from their hours of play and also have real monetization in gaming.

Closure

- It's fair to note that the idea is a growing one and further discussions are still needed to have this ecosystem as effective as possible.
- References:
<https://dev.to/hendrikebbers/gaming-web3-how-an-open-and-secure-future-of-ownership-in-games-could-look-2ihf>
<https://racer.open-elements.cloud/>
<https://dino.open-elements.cloud/>